


Tues Aug 20, 1957

| | |
|---|---|
| Brooks | X Mark Wells |
| Lagomarsino | X Karl Wallick |
| Johnston | X Bill Wood |
| Cooke | X Clark |
| Dave Woods | X |
| Wootton | |
| Voorhees | |
| Kolbeck | |
| Sweeney | |
| Dunwell | |
| Griffith | |

Podin Disc – desire to have presented commercial Problems - so as to increase breadth of L.A. outlook

Dunwell - Cancelled - no time

Factors affecting choice of sample prob

a) Typical & Important

b) 1-8 hrs on Sketch

c) Easily described

d) Possible to declassify (or water down)

e) well understood - numerically

f) numerical data be available or obtainable

Lazarus: there is no sample prob which is typical. A stat. approach must be taken across all probs to get the proper breadth & feeling for the "typical" case.
It is mutually impossible for any one person to suggest alternate ways to do a prob because of the perturbations on other part of the same problem complex. The complex is much too large.

Voorhees - Table of Simple Coding of small parts of probs
Described assumptions in LASL writeup.
No strong feelings abt this proposal - it is for tentative coding only.
V field of Index Reg should line up with LO part of mantissa

Tues PM

Worlton - SNG - assumed no indexing on SA. always used indirect addr on SA to get indexing.
The equations probably will not be altered easily (it is felt) in a manner which would not alter the relative emphasis of the arith operations & control.

Lazarus - SNG is a subroutine embedded in other codes. The matrix of which is a very large complex, and cannot be easily extrapolated from the 704 to the Stretch machine.
It was reported that the largest SNG problem run on the 704 was: J+1=5, I=50, G=25, # iterations unknown, running time = 2 hrs
SNG is a very general code, and therefore probably will remain with us for a long time.

Number size - used halfword on 701 with some trouble

exp underflow gives trouble on 704.

The rate of convergence is controlled by physics - not math - some experimental work could be done on Stretch to improve conv, but probably not much

15 bit mantissa ok for this prob

Output - usually always as input to another prob

Curves occasionally

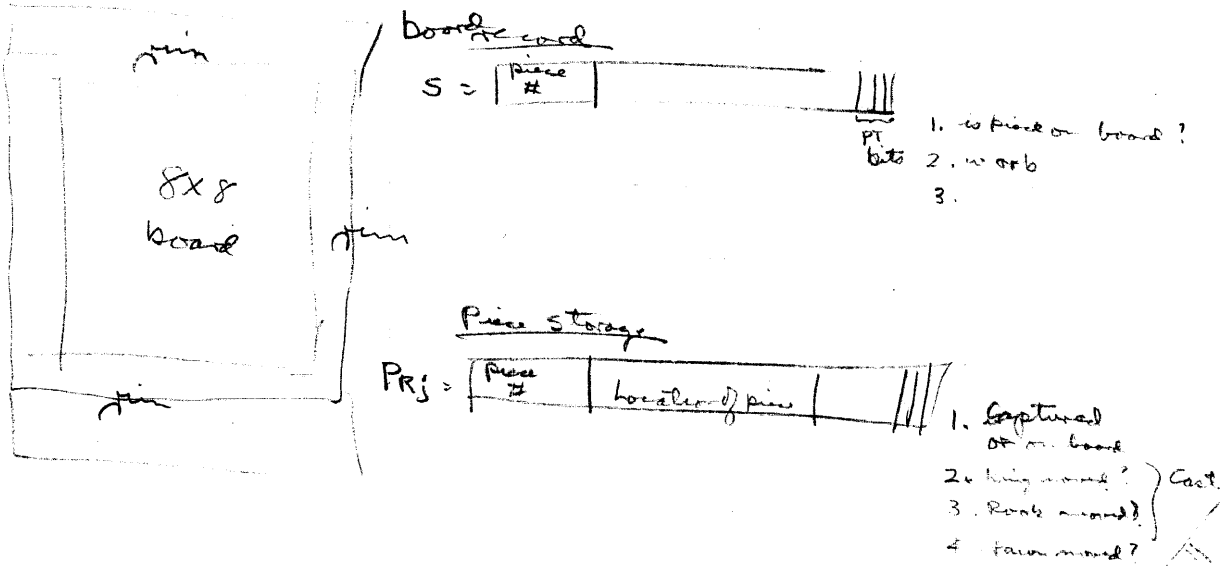
Discussion of possibility of presenting examples of control

Sweeney - SNG - control code of $h_i(g) * \gamma_g f_i \rightarrow S_i(g)$

(Collapsed matrix of cross sections)

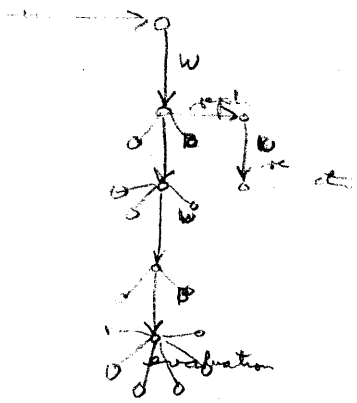
Mark Wells - Chess - 8x8 - pure logic

predictions of all possible continuations of k moves (k/2 wh, k/2 blacks)



Each piece is taken as a subroutine

Example: move by Wh. Kt.

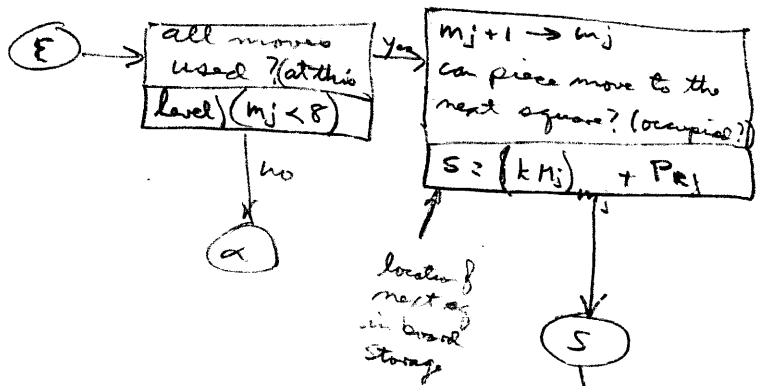



J = move level

$m_j$ = move

$R_j$ =

Piece Storage = $P_{R_j}$




Evaluate at end of each W moves, evaluate B last move, then back up one more move and evaluate W move etc until back at start of 4. Then add one more level and repeat

Indirect addressing very valuable (multi level, too)!

Bit testing is more important!

(6X6 takes on (k=4) maniac: 16 min/move
8X8 on 704 (k=4): 16 min/move)

Non contiguous sets of bits need to be tested!